THANGARAJ T

PYTHON FULL-STACK DEVELOPER | AI / ML ENGINEER

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GitHub: github.com/tmthangaraj1980-AIML | Portfolio: tmthangaraj1980-aiml.github.io/AI-ML-Portfolio/

PROFILE

AI/ML engineer with hands-on Python, FastAPI REST API and JavaScript frontend development. Builds end-to-end AI applications combining retrieval-augmented generation, LangGraph workflows, vector search and practical user interfaces.

FEATURED PROJECT | MEDIGUIDE AI

LangGraph-Orchestrated Healthcare Article RAG Assistant

- Built an interactive article reader and embedded AI assistant with vanilla HTML/CSS/JavaScript and a FastAPI REST backend.
- Ingested 100 MedlinePlus health topics; implemented article-scoped retrieval with Sentence Transformers embeddings and FAISS.
- Integrated Groq LLM and LangGraph orchestration to generate source-grounded answers with article citations and insufficient-evidence handling.
- Delivered a GitHub-hosted codebase and recorded local demonstration; cloud deployment not claimed.

Project: github.com/tmthangaraj1980-AIML/MediGuide-AI

TECHNICAL SKILLS

Python & Backend:	Python, FastAPI, REST APIs, Pydantic, SQL, SQLite, SQLAlchemy
Frontend:	HTML5, CSS3, JavaScript, responsive UI; React fundamentals
Generative AI:	LangGraph, LangChain, RAG, Groq, Gemini, FAISS, Sentence Transformers, Ollama
ML / CV:	scikit-learn, Pandas, NumPy, TensorFlow/Keras, OpenCV, NLP, OCR
Developer Tools:	Git, GitHub, VS Code, Jupyter, virtual environments, Render basics

EXPERIENCE

AI/ML Intern | Femtosoft Technologies, Chennai Sep 2026 - Present

- Developing ClearCustoms AI: OCR/NLP-based customs document extraction, validation and ICEGATE field mapping with Python and FastAPI.

AI/ML Developer Intern | OneYes Infotech, Chennai Aug - Sep 2026

- Built face-recognition attendance workflows using MTCNN and OpenCV, plus document-generation utilities.

AI/ML Intern | DataMites Jan - Jul 2026

- Developed RAG-based PDF chatbot and applied ML projects spanning prediction, NLP and geospatial analysis.

EDUCATION

M.E. Computer Science (AI & ML) - RVS College of Engineering and Technology (in progress)

B.E. Electronics & Communication Engineering - Sri Sairam College of Engineering, Chennai